# EC991 Topics in Macroeconomics

## Week 3 · The Aiyagari Model

Aseem Patel

Department of Economics, University of Essex

# Where we stopped: the sequential problem

- A continuum of households of unit mass; future utility discounted by $\beta \in (0,1)$
- Earnings are $w\,y_t$: the market wage times an idiosyncratic **labour endowment** $y_t$, a Markov chain with transition probabilities $P(y' \mid y) = \Pr\left(y_{t+1} = y' \,\middle|\, y_t = y\right)$
- The only asset is a **riskless bond** paying $r$, with $R = 1 + r$, subject to the borrowing constraint $a_{t+1} \geq -\phi$
- The household's problem, given prices $(w, r)$:

$$\max_{\{c_t,\, a_{t+1}\}} \; \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + a_{t+1} = w\,y_t + R\,a_t, \qquad a_{t+1} \geq -\phi, \quad c_t \geq 0,$$

for $t = 0, 1, \ldots,$ with $a_0$ and $y_0$ given

- Prices carry no time subscript because we look for a **stationary** equilibrium: $w$ and $r$ are constant

# The recursive formulation

- Everything the household needs to know today is its **state** $(a, y)$: assets brought in and current endowment
- The same problem, written recursively (week 2's Bellman equation):

$$V(a, y) \;=\; \max_{a' \geq -\phi} \left\{ u\big(Ra + w\,y - a'\big) \;+\; \beta \sum_{y'} P(y' \mid y)\, V(a', y') \right\}$$

- The solution is a pair of **policy functions**: savings $a' = g_a(a, y)$ and consumption $c = g_c(a, y) = Ra + wy - g_a(a, y)$
- Two properties from week 2 that matter today:
  - the constraint binds at low wealth and low income: there, $g_a(a, y) = -\phi$
  - **precautionary saving**: at any given $r$ the household saves more than it would under certainty

# From partial to general equilibrium

- So far this is a **partial equilibrium** exercise: one household, prices fixed
- In **general equilibrium** the interest rate is the price that makes households' savings equal what the economy needs: $r$ must clear the asset market
- We look for a **stationary equilibrium**: aggregates (total assets, output) and prices are constant through time, while every individual household keeps moving up and down the earnings and wealth distribution
- Such an equilibrium has a **stationary distribution** of households over $(a, y)$: a distribution that replicates itself period after period
- Today: the stationary distribution and asset supply; closing the model with firms; existence; what the model delivers; then the first steps of computing it

# Where households are: the distribution

- A household is characterised by its pair $(a, y)$: assets and endowment
- The **distribution** at date $t$ records what share of households sits at each pair:

$$\lambda_t(a, y) \;=\; \Pr\left(a_t = a,\; y_t = y\right)$$

- **Running example**: 100 workers, two asset levels $a \in \{1, 2\}$ and two endowments $y \in \{1, 2\}$, with 1 low and 2 high. Shares at date $t$, workers in brackets:

date $t$

| | $y = 1$ | $y = 2$ |
|---|---|---|
| $a = 1$ | 0.40 (40) | 0.20 (20) |
| $a = 2$ | 0.15 (15) | 0.25 (25) |

$\Longrightarrow$

date $t + 1$

| | $y = 1$ | $y = 2$ |
|---|---|---|
| $a = 1$ | ? | ? |
| $a = 2$ | ? | ? |

- The table **is** $\lambda_t$: the highlighted cell says $\lambda_t(1, 1) = \Pr(a_t = 1,\, y_t = 1) = 0.40$, i.e. 40 of the 100 workers
- **The question**: given where households are today, where are they tomorrow?

# Where can one household go next?

- A household at $(a, y)$ moves through two separate channels:
  - its **savings choice** fixes tomorrow's assets: $a' = g_a(a, y)$. No randomness here
  - the **endowment chain** draws tomorrow's income: $y'$ with probability $P(y' \mid y)$

- The probability of reaching **one particular destination** $(a', y')$ is the product of the two:

$$\underbrace{Q\big((a,y),\ \{(a',y')\}\big)}_{\text{probability of reaching this destination}} = \underbrace{\mathcal{I}\{g_a(a,y) = a'\}}_{\text{does the asset choice match? 1 if yes, 0 if no}} \times \underbrace{P(y' \mid y)}_{\text{probability of this endowment}}$$

- **Example**
  - Assume the savings policy: save when income is high, $g_a(a, 2) = 2$; dissave when it is low, $g_a(a, 1) = 1$
  - Assume the endowment chain: $P(2 \mid 1) = P(1 \mid 2) = 0.2$
  - Take a worker at $(1, 2)$, poor with high income. It saves, $a' = g_a(1, 2) = 2$, and then draws its income:

$$(1,2) \xrightarrow{a'=2} \begin{cases} (2,1) & \text{with probability } P(1 \mid 2) = 0.2 \\ (2,2) & \text{with probability } P(2 \mid 2) = 0.8 \end{cases}$$

  - It reaches $(1, 1)$ and $(1, 2)$ with probability 0: the asset choice does not match

# Transitions

- For any group of destinations $\mathcal{A} \times \mathcal{Y}$: check whether the chosen assets belong to the group, then add up the probabilities of the endowments in it. That is the **transition function**:

$$Q\big((a, y),\, \mathcal{A} \times \mathcal{Y}\big) \;=\; \mathcal{I}\{g_a(a, y) \in \mathcal{A}\} \sum_{y' \in \mathcal{Y}} P(y' \mid y)$$

- **Example**
  - Assets $A = \{1, 2\}$ and endowment $Y = \{1, 2\}$, so tomorrow the household is in one of **four** states: $A \times Y = \big\{(1,1),\ (1,2),\ (2,1),\ (2,2)\big\}$
  - A household at $(1, 1)$ has low income, so it dissaves, $g_a(1, 1) = 1$. For each of the four states:

| destination $(a', y')$ | $(1,1)$ | $(1,2)$ | $(2,1)$ | $(2,2)$ |
|---|---|---|---|---|
| $\mathcal{I}\{g_a(1,1) = a'\}$ | 1 | 1 | 0 | 0 |
| $P(y' \mid 1)$ | 0.8 | 0.2 | 0.8 | 0.2 |
| $Q\big((1,1), (a', y')\big)$ | **0.8** | **0.2** | **0** | **0** |

  - So $Q\big((1,1), \cdot\big)$ gives the **probability of each of the four states** the household can be in tomorrow; the four numbers sum to one

# Stationary distribution

- To get next period's distribution, apply the transition function $Q$ to every point of today's distribution:

$$\Lambda_{t+1}(\mathcal{A} \times \mathcal{Y}) = \int_{A \times Y} \underbrace{Q((a,y),\, \mathcal{A} \times \mathcal{Y})}_{\text{probability of reaching the destination set}} \underbrace{d\Lambda_t(a,y)}_{\text{population weight at the starting state}}$$

- This is the **law of total probability** applied to the population: tomorrow's share in a destination set is the sum of arrivals from every possible starting state

| | |
|---|---|
| $A \times Y$ | the **entire state space**, over which today's households are distributed |
| $\mathcal{A} \times \mathcal{Y}$ | the **destination set** whose population share tomorrow we want |
| $\Lambda_{t+1}(\mathcal{A} \times \mathcal{Y})$ | the fraction of households in that destination set tomorrow |

- We hold the **destination set fixed** and integrate over **all starting states**. $d\Lambda_t$ means "weight by today's distribution"; it is not a change in the distribution over time
- The **stationary distribution** is the one that replicates itself: for all $(a, y)$, $\lambda(a,y) = \lambda_t(a,y) = \lambda_{t+1}(a,y)$
  - households still move between states; at every state, arrivals and departures balance

# The law of motion in the example

- With four states, the population weight at a starting state is just its cell in today's table: $d\Lambda_t(a, y)$ becomes $\lambda_t(a, y)$, and the integral becomes a **sum over the four cells**:

$$\underbrace{\Lambda_{t+1}(\mathcal{A} \times \mathcal{Y})}_{\text{tomorrow}} = \sum_{a \in \{1,2\}} \sum_{y \in \{1,2\}} Q\big((a, y),\, \mathcal{A} \times \mathcal{Y}\big)\, \underbrace{\lambda_t(a, y)}_{\text{today}}$$

- $\mathcal{A} \times \mathcal{Y}$ is a **generic destination set**. We now choose it in two different ways:
- **One cell**: $\mathcal{A} = \{1\}$, $\mathcal{Y} = \{1\}$, so $\Lambda_{t+1}\big(\{1\} \times \{1\}\big) = \lambda_{t+1}(1, 1)$:

$$\lambda_{t+1}(1,1) = \underbrace{0.8 \times 0.40}_{\text{from (1,1)}} + \underbrace{0 \times 0.20}_{\text{from (1,2)}} + \underbrace{0.8 \times 0.15}_{\text{from (2,1)}} + \underbrace{0 \times 0.25}_{\text{from (2,2)}} = \mathbf{0.44}$$

- **Two cells**: $\mathcal{A} = \{1\}$, $\mathcal{Y} = \{1, 2\}$ selects low assets and either income:

$$\begin{aligned} \Lambda_{t+1}\big(\{1\} \times \{1,2\}\big) &= \lambda_{t+1}(1,1) + \lambda_{t+1}(1,2) \\ &= 0.44 + \underbrace{0.2 \times (0.40 + 0.15)}_{\text{share reaching (1,2)}} = \mathbf{0.55} \end{aligned}$$

- So 55% have low assets tomorrow: we add the shares in the two destination cells

# The law of motion in matrix form

- The matrix $Q$ governs how the fraction in each state today is spread over the states tomorrow:

$$\underbrace{\begin{bmatrix} \lambda_{t+1}(a_1, y_1) \\ \lambda_{t+1}(a_1, y_2) \\ \lambda_{t+1}(a_2, y_1) \\ \lambda_{t+1}(a_2, y_2) \end{bmatrix}}_{\lambda_{t+1}} = \underbrace{\begin{bmatrix} Q_{11} & Q_{21} & Q_{31} & Q_{41} \\ Q_{12} & Q_{22} & Q_{32} & Q_{42} \\ Q_{13} & Q_{23} & Q_{33} & Q_{43} \\ Q_{14} & Q_{24} & Q_{34} & Q_{44} \end{bmatrix}}_{Q'} \underbrace{\begin{bmatrix} \lambda_t(a_1, y_1) \\ \lambda_t(a_1, y_2) \\ \lambda_t(a_2, y_1) \\ \lambda_t(a_2, y_2) \end{bmatrix}}_{\lambda_t}$$

  where $Q_{mn}$ is the probability of moving from state $m$ to state $n$, so $\lambda_{t+1} = Q'\lambda_t$ (rows of $Q$ sum to one)

- Under certain conditions the Markov chain admits a **unique** stationary distribution $\lambda(a, y)$

# Asset supply

- With the stationary distribution in hand, aggregate asset supply is the sum of the savings of all households
- That is, we integrate the savings policy against the distribution to get the **asset supply function**:

$$\mathbb{E}a(r) \;=\; \int_{A\times Y} g_a(a, y;\; r)\; d\Lambda(a, y;\; r)$$

- $r$ enters twice: through the savings policy, and through the distribution itself
  - a change in $r$ changes households' savings choices **and** the stationary distribution those choices generate
  - to obtain asset supply at another $r$, we must recompute **both**
- A higher interest rate increases the incentive to **postpone consumption**. Its effect on stationary aggregate asset holdings also depends on **income effects** and on changes in the **distribution**

# Closing the model

To close the model, we must define the **demand for assets** in the economy. Two options:

- **Huggett (1993): credit economy.** Some agents borrow, others lend; the loan market clears when the aggregate demand for loans is zero:

$$\int_{A\times Y} g_a(a,y)\, d\Lambda \;=\; 0$$

- **Aiyagari (1994): production economy.** Firms demand capital to produce; the market clears when household savings equal capital demand:

$$\int_{A\times Y} g_a(a,y)\, d\Lambda \;=\; K$$

- We follow Aiyagari (1994) and assume an aggregate production function

# Firms

- Production function: $Y = F(K, N) = K^{\alpha} N^{1-\alpha}$, with $\alpha \in (0, 1)$
- Capital depreciates at rate $\delta$
- Markets are competitive; the firm's problem is standard (time subscripts omitted):

$$w = \frac{\partial F(K,N)}{\partial N} = (1-\alpha)\left(\frac{K}{N}\right)^{\alpha}, \qquad r + \delta = \frac{\partial F(K,N)}{\partial K} = \alpha\left(\frac{K}{N}\right)^{-(1-\alpha)}$$

- A **tight connection** between $w$ and $r$ through the capital-labour ratio: $\uparrow r \Leftrightarrow \downarrow K/N \Leftrightarrow \downarrow w$
  - a higher interest rate makes capital more expensive, so firms use **less capital per worker**
  - with less capital to work with, workers are less productive, and competitive **wages are lower**

# Labour supply: the stationary endowment distribution

- There is a **continuum of households of mass one**, each supplying its endowment $y$ as effective labour
- Endowments evolve according to the **exogenous** transition matrix $P$
- Let $\pi_j$ be the stationary population share with endowment $y_j$. The vector $\pi$ satisfies $\pi = P'\pi$, $\sum_j \pi_j = 1$
- **Two distributions**: $\pi$ describes endowments $y$; $\Lambda$ describes joint states $(a, y)$
  - in stationary equilibrium they agree on endowment shares: $\pi_j = \Lambda\big(A \times \{y_j\}\big)$, counting households with endowment $y_j$ **across all asset holdings**
- Since labour supplied depends only on $y$, aggregate effective labour is

$$N = \sum_j y_j \pi_j$$

- **Example**: with the symmetric $P$, half have $y = 1$ and half have $y = 2$, so $N = 1 \times 0.5 + 2 \times 0.5 = \mathbf{1.5}$
- When we vary $r$, $P$ and its unique stationary distribution $\pi$ remain fixed. Hence $N$ **remains fixed**, although savings choices and the joint distribution change

# Equilibrium definition

A **stationary recursive competitive equilibrium** is a value function $V$; household policy functions $g_a$ and $g_c$; firm choices $K$ and $N$; prices $w$ and $r$; and a stationary distribution $\Lambda$ such that:

1. Given prices, $V$, $g_a$ and $g_c$ solve the **household problem**
2. Given prices, $K$ and $N$ solve the **firm's problem**
3. Given the transition function $Q$, the stationary distribution satisfies $\Lambda(\mathcal{A} \times \mathcal{Y}) = \int Q\big((a, y), \mathcal{A} \times \mathcal{Y}\big)\, d\Lambda$
4. The **labour market** clears: $N = \sum_j y_j \pi_j$
5. The **asset market** clears: $\int g_a(a, y)\, d\Lambda = K$
6. The **goods market** clears: $\int g_c(a, y)\, d\Lambda + \delta K = F(K, N)$

# Equilibrium in one unknown: the interest rate

**The test**: for each candidate interest rate $r$, compare the capital firms demand with the assets households supply

$$Z(r) \;\equiv\; K(r) - \mathbb{E}a(r)$$

- $Z(r) > 0$: firms want more capital than households supply
- $Z(r) < 0$: households supply more assets than firms want to use
- $Z(r) = 0$: the plans match, and the asset market clears. **An equilibrium is an $r^*$ with $Z(r^*) = 0$**

- **One price is enough**: $N$ is fixed by the endowment process. Given $N$, each $r$ fixes the firm's capital demand, and the resulting $K/N$ fixes the wage:

$$K(r) \;=\; N\left(\frac{\alpha}{r+\delta}\right)^{\frac{1}{1-\alpha}}, \qquad w(r) \;=\; (1-\alpha)\left(\frac{K(r)}{N}\right)^{\alpha}$$

- **Asset supply at each $r$**: solve the household problem at **both** prices $\big(r, w(r)\big)$, find the stationary distribution $\Lambda$, and aggregate to $\mathbb{E}a(r)$

- **To show an $r^*$ exists**: $Z$ continuous, with a **sign change** between two interest rates

# Households' desired wealth

**Households' supply**: we have firms' demand $K(r)$. Now we need asset supply $\mathbb{E}a(r)$, particularly near the highest sustainable return

- **Income risk encourages saving**: wealth protects consumption against bad income shocks
- **Protection has diminishing benefits**: another unit of wealth matters less when the buffer is already large
- **Impatience encourages consumption today**: when $\beta R < 1$, the return does not fully compensate for waiting
- **A finite buffer balances these forces**: eventually, the extra protection from one more unit of wealth is too small to justify postponing more consumption
- **As $r$ rises**, waiting is better rewarded, so households are willing to hold **larger buffers**
- **As** $\beta R \to 1$: ever larger buffers are needed before the extra protection is small enough to stop saving

Under the standard Aiyagari assumptions:

$$r \to \bar{r} \equiv \frac{1}{\beta} - 1 \qquad \Longrightarrow \qquad \mathbb{E}a(r) \to +\infty$$

Capital demand $K(r)$ remains finite, so near this boundary there is **excess asset supply**:

$$Z(r) = K(r) - \mathbb{E}a(r) < 0 \qquad \text{for } r \text{ sufficiently close to } \bar{r}$$

# Finding an interest rate that clears the asset market

- **At sufficiently low interest rates**: capital is cheap for firms, while saving offers households a low return. Firms demand more capital than households supply:

$$Z(r_L) \;=\; K(r_L) - \mathbb{E}a(r_L) \;>\; 0$$

- **Near the upper boundary**: from the previous slide, household asset supply exceeds firms' capital demand:

$$Z(r_H) \;=\; K(r_H) - \mathbb{E}a(r_H) \;<\; 0$$

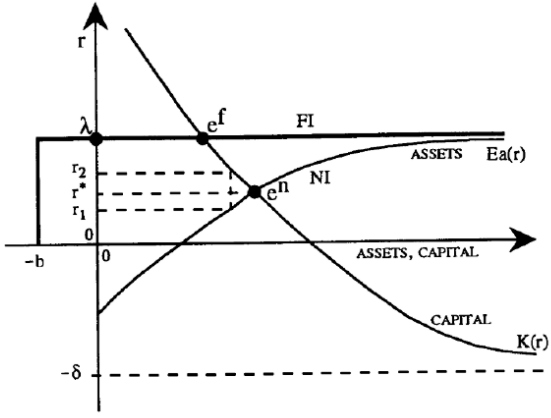
- **If excess demand changes continuously**: somewhere between these rates, household savings and firms' capital demand match:

$$K(r^*) \;=\; \mathbb{E}a(r^*)$$

- At least **one** equilibrium exists; there may be more than one

More details

# General equilibrium: where supply meets demand




Aiyagari 1994, Figure IIb. Illustrative curve shapes; uniqueness is not guaranteed.

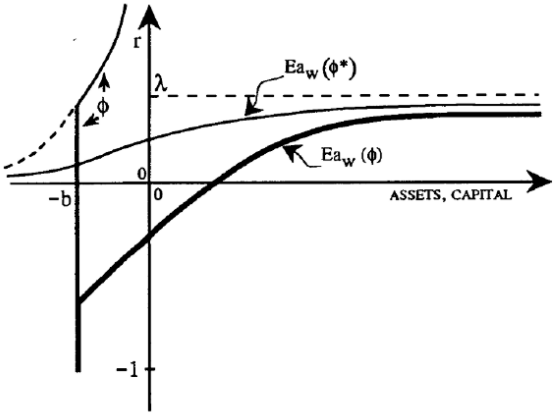
**Curves**
$K(r)$: firms' capital demand.
$\mathbb{E}a(r)$: household asset supply.
NI: no insurance against income risk.

**Incomplete-markets equilibrium**
At $e^n$: $\mathbb{E}a(r^*) = K(r^*)$.
Below $r^*$, at $r_1$: excess demand.
Above $r^*$, at $r_2$: excess supply.

**Full-insurance benchmark**
FI: full insurance; equilibrium at $e^f$.
At $e^n$: more capital and a lower $r$.

**Original notation**
$\lambda = \bar{r} = 1/\beta - 1$: a rate,
not the household distribution.

# Asset supply: the role of borrowing limits




Aiyagari 1994, Figure IIa. For $r > 0$, the natural debt limit is $\phi^* = wy_{\min}/r$.

**Two borrowing restrictions**
Thick curve, $\mathbb{E}a_w(\phi)$:
tighter borrowing limit.
Thin curve, $\mathbb{E}a_w(\phi^*)$:
borrowing up to the natural debt limit.

**Read horizontally**
In the comparison shown, more borrowing flexibility means fewer assets.

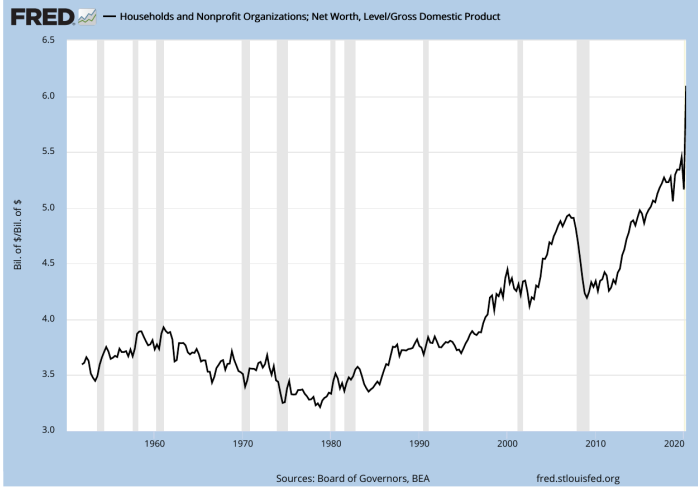
**Intuition**
Borrowing can help cover low income.
Households rely less on saved assets.

**Original notation**
$w$ in $\mathbb{E}a_w$: wage held fixed.
$-b$: imposed asset floor (our $-\phi$).
$\lambda = \overline{r} = 1/\beta - 1$.

# Calibration

- **Standard parameters**: $\alpha = 0.33$, CRRA utility with $\gamma$ between **1 and 5**, $\delta = 0.08$
- **Discount factor** $\beta$: chosen to match the aggregate or average **wealth-to-income ratio**
  - option 1, **macro target**: aggregate household wealth over GDP from the national accounts; for the U.S. about **3 to 5**, depending on the period and on whether residential capital is included
  - option 2, **micro target**: average wealth from a household survey, such as the Survey of Consumer Finances (SCF)
- **Labour income process**: estimated from micro data on individual income dynamics, as an AR(1) $\log y_t = \rho \log y_{t-1} + \sigma \varepsilon_t$, then discretised to a Markov chain (this week's tutorial)
- **Borrowing limit** $\phi$: chosen to match the fraction of households with **negative net worth**, around **10%** in the U.S. (or set $\phi = 0$)

# Calibration target: aggregate wealth-to-GDP ratio, U.S.




- Household net worth relative to GDP stayed between about **3** and **4** until the mid-1990s and has been between about **4** and **5.5** since: the value of the target depends on the period

**FRED: Households and Nonprofit Organizations; Net Worth, Level / Gross Domestic Product (Board of Governors, BEA)**

# Calibration target: average household wealth (SCF 2016)

| Wealth definition | | All | Excluding top 1% |
|---|---|---|---|
| Total | Mean | 11.2 | 7.8 |
| | Median | 1.6 | 1.7 |
| Non-housing | Mean | 8.5 | 5.3 |
| | Median | 0.5 | 0.5 |
| Financial | Mean | 5.3 | 4.0 |
| | Median | 0.3 | 0.3 |
| Mean earnings | | $61,600 | $54,900 |
| Mean income | | $102,200 | $84,300 |

- Wealth is expressed as a ratio to mean household **earnings**
- Average wealth $= 11.2 \times \$61{,}600 = \$689{,}920$
- Average wealth / average income $= \$689{,}920/\$102{,}200 =$ **6.75**
  - surprisingly hard to square with the aggregate ratio on the previous slide

Survey of Consumer Finances 2016

# Calibration target: low-wealth households (SCF 2016)

| Wealth definition | | Share of households |
|---|---|---|
| Total | $\leq 0$ | 11% |
| | $\leq$ \$2,000 | 17% |
| Non-housing | $\leq 0$ | 15% |
| | $\leq$ \$2,000 | 22% |
| Financial | $\leq 0$ | 11% |
| | $\leq$ \$2,000 | 31% |

- About **11%** of households have zero or negative total net worth: the target for the borrowing limit $\phi$
- Many more hold almost nothing: **31%** have \$2,000 or less in financial wealth

Survey of Consumer Finances 2016

# Why does precautionary saving raise the saving rate only modestly?

**Result (Aiyagari 1994)**: for reasonable calibrations, the difference between saving rates under complete and incomplete markets is **small**, at most **2%**

**A force and a counteracting feedback**

- **The force**: uninsured households want **larger asset buffers** to protect consumption against bad income shocks
- **The feedback**: the additional assets finance additional capital. More capital per worker lowers its marginal product, so the interest rate **falls**
- **The consequence**: the lower return discourages some of the extra accumulation households would have wanted at unchanged prices, so the **equilibrium** increase in saving is limited

**When can the difference be larger?**

- With much larger or more persistent income shocks, the precautionary motive is stronger, so the saving-rate difference can be **substantially larger**
  - **larger** $\sigma$: bad shocks are deeper, so households need a larger buffer
  - **larger** $\rho$: low income lasts longer, so savings must cover a longer spell

Aiyagari 1994, p. 672

# Why wealth > income > consumption inequality

**Result (Aiyagari 1994)**: inequality ranks as in the data: **wealth** > **income** > **consumption**

Households transfer resources across dates: this **cushions consumption** and carries **earnings histories** into wealth

**Consumption responds less to temporary income differences**

- Concave utility encourages households to spread consumption across dates
- They save part of favourable earnings and draw down assets or borrow after adverse shocks
- Borrowing constraints restrict smoothing; more persistent earnings shocks require larger consumption adjustments

# Why wealth > income: asset differences outlast earnings differences

- Exogenous earnings histories differ; optimal saving decisions translate them into asset holdings
- Current earnings can become similar again while accumulated assets remain different
- Households with little wealth still earn wages: when earnings are less concentrated than assets, total income is less unequal than wealth

| At the comparison date | Household A | Household B |
|---|---|---|
| Past earnings | Favourable history | Adverse history |
| Current earnings | Same | Same |
| Expected future labour earnings | Same | Same |
| Accumulated assets | **Larger buffer** | **Smaller buffer** |

# Wealth is right-skewed, but not concentrated enough

**Result (Aiyagari 1994)**: the wealth distribution is **right-skewed**, so top inequality is a feature of these models

**But the baseline is quantitatively off**

- Wealth inequality in the model is **lower** than in the data
- In the SCF, mean wealth is **11.2** times mean earnings, while median wealth is only **1.6** times: a few households hold very large fortunes

**Why: limited incentives to retain enormous fortunes**

- Once wealth provides substantial protection against earnings risk, additional insurance is less valuable
- With $\beta(1+r) < 1$, impatience then encourages wealthy households to **consume their assets**

# Conclusion

- The incomplete-markets model delivers **new theoretical insights** and reopens old questions: capital taxation, government debt, …
- But where the model **shines** is as a framework for new questions about **income and wealth inequality**
- A large subsequent literature works to make the model **match the wealth distribution** well
- And then studies policies where inequality is central: progressive taxation, social security, …

# Where to go now?

- **Progressive taxation**: Boar and Midrigan (2022, JME); Kindermann and Krueger (2022, AEJ: Macro); Brüggemann (2021, AEJ: Macro); Heathcote and Tsujiyama (2021, JPE); Conesa, Kitao and Krueger (2009, AER)
- **Hours and labour market frictions**: Bick, Blandin and Rogerson (2022, QJE); Krusell, Mukoyama and Şahin (2010, ReStud)
- **Top wealth inequality and heterogeneous returns**: De Nardi and Fella (2017, RED); Benhabib, Bisin and Luo (2019, AER); Benhabib, Bisin and Zhu (2011, ECTA); Kaymak, Leung and Poschke (2022)
- **Liquid and illiquid assets, the wealthy hand-to-mouth, MPCs**: Kaplan and Violante (2014, ECTA; 2022, Annual Review)
- **Unsecured credit and consumer default**: Chatterjee et al. (2007, ECTA); Livshits et al. (2007, AER); Dempsey and Ionescu (2022); Herkenhoff (2019, ReStud)
- **Non-homothetic preferences**: Straub (2022); Mian, Straub and Sufi (2021, QJE); Carroll and Hur (2020, JME); Känzig (2022)

# Extension: Fiscal Policy

- Households face uninsured earnings risk and use assets to protect consumption
- Fiscal policy changes households' resources, saving opportunities and equilibrium prices
- **Can these changes improve welfare after accounting for their financing and production costs?**

**Introduce government debt**

Households' financial assets now finance both productive capital and government debt:

$$\boxed{A = K + B}$$

- $A$: assets held by households
- $K$: productive capital
- $B$: government debt held by households
- **Household assets and productive capital are no longer the same object**

# How taxes and transfers enter household decisions

The household budget becomes:

$$(1+\tau_c)\, c_t + a_{t+1} \;=\; (1-\tau_w)\, w y_t + \left[1 + r(1-\tau_r)\right] a_t + \tau$$

- **Labour-income tax** $\tau_w$: households retain less of their earnings
- **Capital-income tax** $\tau_r$: households retain less of the return to saving
- **Consumption tax** $\tau_c$: consumption costs more
- **Transfer** $\tau$: households receive additional resources

**Intuition**

- Transfers can support consumption when earnings or assets are low
- Financing those transfers changes disposable resources and potentially saving incentives
- Households choose consumption and saving taking the **entire tax-and-transfer system** into account

# Government debt must be financed

The government pays for purchases and debt repayment using net taxes and new borrowing:

$$G_t + (1 + r)B_t \;=\; B_{t+1} + T_t^{\text{net}}$$

In steady state, debt is constant: new borrowing replaces maturing principal. Therefore,

$$\boxed{G + rB \;=\; \tau_w\, wN + \tau_r\, rA + \tau_c\, C - \tau}$$

- Taxes finance government purchases, interest payments and transfers
- Transfers subtract from net revenue
- **More debt creates an interest bill that must be financed**

**Our policy experiment**

- Increase $B$, keeping $G$, $\tau$, $\tau_c$, $\tau_r$ fixed
- Adjust $\tau_w$ so the government budget holds
- Because labour supply is fixed here, this tax adjustment changes disposable earnings, **not hours worked**

# More debt: what happens to capital and returns?

Household asset holdings respond to the new policy. Market clearing requires:

$$K = A - B$$

- If household assets increase by less than public debt, **productive capital falls**
- With less capital per worker, capital's marginal product rises: $K/N \downarrow \implies r \uparrow,\ w \downarrow$
- Lower capital also means lower output

**Intuition**

- Some household wealth now finances government debt instead of productive capital
- **Capital can fall even if households hold more financial assets**
- Government bonds provide the same risk-free saving opportunity as claims on capital; their importance comes from changing equilibrium quantities and returns
- These are equilibrium responses, not a claim that each additional unit of debt mechanically eliminates one unit of capital

# Does the policy improve welfare?

**Potential benefit**

- With the capital-income tax rate fixed, a higher interest rate raises the after-tax reward to saving
- This can encourage larger private asset buffers
- Larger buffers can help households maintain consumption after adverse earnings shocks

**Costs**

- Lower capital reduces wages and output
- Debt service requires financing through the government budget
- The resulting tax adjustment changes households' disposable earnings and saving decisions

# Does the policy improve welfare? Weighing the two

**The question**: do improved self-insurance opportunities outweigh the losses from lower production and the distributional effects of financing?

- Larger buffers are a possible outcome, not an automatic consequence of issuing bonds
- A welfare improvement under a chosen social criterion need not benefit every household
- Aiyagari–McGrattan find positive optimal debt and **small** gains from moving from existing debt to the optimum
- Life-cycle extensions can instead favour **public asset accumulation**

Aiyagari and McGrattan (1998, JME); Peterman and Sager (2022, AEJ: Macro)

# Backup: excess demand at low interest rates

- As $r \to -\delta$, capital costs $r + \delta \to 0$ per unit, so $K(r) \to +\infty$
- This alone does not sign $Z$: the wage $w(r)$ also rises without bound, and richer households may save more
- So compare the two sides **in units of the wage**:

$$\frac{K(r)}{w(r)} = \frac{\alpha}{1-\alpha}\frac{N}{r+\delta} \to +\infty$$

- When $\beta R < 1$, households hold a finite buffer, so $\mathbb{E}a(r)/w(r)$ stays bounded under the maintained assumptions
- Hence $Z(r) > 0$ for $r$ sufficiently close to $-\delta$

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